

2009 年数学(三)真题解析

一、选择题

(1)【答案】 (C).

【解】 $x=k(k \in \mathbf{Z})$ 为函数 $f(x)$ 的间断点,

$$\text{由 } \lim_{x \rightarrow 0} f(x) = \frac{1}{\pi},$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x(1-x)(1+x)}{\sin[\pi(1-x)]} = \frac{2}{\pi},$$

$$\lim_{x \rightarrow -1} f(x) = - \lim_{x \rightarrow -1} \frac{x(1-x)(1+x)}{\sin[\pi(1+x)]} = \frac{2}{\pi},$$

得 $x=0, 1, -1$ 为 $f(x)$ 的可去间断点,

当 $x=k(k \in \mathbf{Z})$ 且 $x \neq 0, 1, -1$ 时, $\lim_{x \rightarrow k} f(x) = \infty$, 故 $f(x)$ 只有 3 个可去间断点, 选(C).

(2)【答案】 (A).

【解】 因为 $g(x) \sim -bx^3$ 为 3 阶无穷小, 所以 $a=1$,

$$x - \sin x = x - \left[x - \frac{x^3}{6} + o(x^3) \right] \sim \frac{1}{6}x^3,$$

由 $f(x) \sim g(x)$ 得 $b = -\frac{1}{6}$, 应选(A).

(3)【答案】 (A).

【解】 令 $f(x) = \int_1^x \frac{\sin t}{t} dt - \ln x (x > 0)$,

$$\text{则 } f(x) = \int_1^x \frac{\sin t}{t} dt - \int_1^x \frac{1}{t} dt = \int_1^x \frac{\sin t - 1}{t} dt = \int_x^1 \frac{1 - \sin t}{t} dt,$$

显然当 $0 < x < 1$ 时, $f(x) > 0$; 当 $x \geq 1$ 时, $f(x) \leq 0$, 故 x 的范围为 $(0, 1)$, 选(A).

(4)【答案】 (D).

【解】 当 $-1 \leq x \leq 0$ 时, $f(x) = 1$, 则 $F(x) = \int_0^x f(t) dt = x$, 排除(A), (C);

因为 $f(x)$ 在 $[-1, 3]$ 上只有一个第一类间断点, 所以 $F(x)$ 为连续函数, 排除(B), 应选(D).

(5)【答案】 (B).

【解】 $\begin{vmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{vmatrix} = (-1)^{2 \times 2} |\mathbf{A}| \cdot |\mathbf{B}| = 6$, 则

$$\begin{aligned} \begin{pmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{pmatrix}^* &= \begin{vmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{vmatrix} \begin{pmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{pmatrix}^{-1} = 6 \begin{pmatrix} \mathbf{O} & \mathbf{B}^{-1} \\ \mathbf{A}^{-1} & \mathbf{O} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{O} & 6\mathbf{B}^{-1} \\ 6\mathbf{A}^{-1} & \mathbf{O} \end{pmatrix} = \begin{pmatrix} \mathbf{O} & 2\mathbf{B}^* \\ 3\mathbf{A}^* & \mathbf{O} \end{pmatrix}, \end{aligned}$$

应选(B).

方法点评: 本题考查逆矩阵与伴随矩阵的性质.

设 $\mathbf{A}$ 为 n 阶可逆矩阵, 涉及 $\mathbf{A}^*$ 时注意使用如下性质:

(1) $AA^* = A^*A = |A|E$, 尤其 $A = |A|A^{-1}$, 当 A 可逆时, A^* 的问题往往转化为 A^{-1} .

(2) 设 A, B 为 n 阶可逆矩阵, 则 $(AB)^* = |AB| \cdot (AB)^{-1} = B^*A^*$.

(3) 设 A, B 分别为 m 与 n 阶可逆矩阵, 则

$$\begin{pmatrix} A & O \\ O & B \end{pmatrix}^* = \begin{vmatrix} A & O \\ O & B \end{vmatrix} \cdot \begin{pmatrix} A & O \\ O & B \end{pmatrix}^{-1} = \begin{pmatrix} |B|A^* & O \\ O & |A|B^* \end{pmatrix};$$

$$\begin{pmatrix} O & A \\ B & O \end{pmatrix}^* = \begin{vmatrix} O & A \\ B & O \end{vmatrix} \cdot \begin{pmatrix} O & A \\ B & O \end{pmatrix}^{-1} = (-1)^{mn} \begin{pmatrix} O & |A|B^* \\ |B|A^* & O \end{pmatrix}.$$

(6)【答案】 (A).

【解】 由 $Q = (\alpha_1 + \alpha_2, \alpha_2, \alpha_3) = P \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$,

得 $Q^T A Q = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^T A P \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

应选(A).

(7)【答案】 (D).

【解】 若 A 与 B 互不相容, 则 $AB = \emptyset$,

于是 $P(\overline{A \cup B}) = P(\overline{AB}) = 1 - P(AB) = 1$, 应选(D).

(8)【答案】 (B).

【解】 $F_Z(z) = P\{Z \leq z\} = P\{XY \leq z\}$

$$= P\{Y=0\}P\{XY \leq z \mid Y=0\} + P\{Y=1\}P\{XY \leq z \mid Y=1\}$$

$$= P\{Y=0\}P\{z \geq 0\} + P\{Y=1\}P\{X \leq z\}$$

$$= \frac{1}{2}P\{z \geq 0\} + \frac{1}{2}P\{X \leq z\},$$

当 $z < 0$ 时, $F_Z(z) = \frac{1}{2}\Phi(z)$;

当 $z \geq 0$ 时, $F_Z(z) = \frac{1}{2} + \frac{1}{2}\Phi(z)$,

即 $F_Z(z) = \begin{cases} \frac{1}{2}\Phi(z), & z < 0, \\ \frac{1}{2} + \frac{1}{2}\Phi(z), & z \geq 0, \end{cases}$ 显然 $F_Z(z)$ 只有一个间断点, 应选(B).

二、填空题

(9)【答案】 $\frac{3e}{2}$.

【解】 由 $\sqrt[3]{1+x^2} - 1 = (1+x^2)^{\frac{1}{3}} - 1 \sim \frac{x^2}{3}$,

$$\begin{aligned}\text{得 } \lim_{x \rightarrow 0} \frac{e - e^{\cos x}}{\sqrt[3]{1+x^2} - 1} &= 3 \lim_{x \rightarrow 0} \frac{e - e^{\cos x}}{x^2} = -3 \lim_{x \rightarrow 0} \frac{e^{\cos x-1} - 1}{x^2} \\ &= -3 \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = 3 \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{3e}{2}.\end{aligned}$$

(10)【答案】 $2\ln 2 + 1$.

【解】 $z = (x + e^y)^x = e^{x \ln(x + e^y)}$,

$$\text{则 } \frac{\partial z}{\partial x} = e^{x \ln(x + e^y)} \cdot \left[\ln(x + e^y) + \frac{x}{x + e^y} \right],$$

$$\text{于是 } \left. \frac{\partial z}{\partial x} \right|_{(1,0)} = e^{\ln 2} \cdot \left(\ln 2 + \frac{1}{2} \right) = 2\ln 2 + 1.$$

(11)【答案】 $\frac{1}{e}$.

$$\begin{aligned}\text{【解】 由 } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{\frac{e^{n+1} - (-1)^{n+1}}{(n+1)^2}}{\frac{e^n - (-1)^n}{n^2}} = \lim_{n \rightarrow \infty} \frac{e^{n+1} - (-1)^{n+1}}{e^n - (-1)^n} \\ &= \lim_{n \rightarrow \infty} \frac{1 - \frac{(-1)^{n+1}}{e^{n+1}}}{1 - \frac{(-1)^n}{e^n}} = e,\end{aligned}$$

得幂级数的收敛半径为 $R = \frac{1}{e}$.

(12)【答案】 8 000.

【解】 收益函数为 $L(P) = PQ$,

所求的量为 $L'(P) = Q + PQ'$,

$$\text{因为 } \frac{\frac{dQ}{dP}}{\frac{Q}{P}} = \frac{PQ'}{Q} = -0.2, \text{ 所以 } PQ' = -0.2Q,$$

于是 $L'(P) = 0.8Q$,

将 $Q = 10\ 000$ 代入, 得 $L'(P) = 8\ 000$, 即价格每增加 1 元会使产品收益增加 8 000 元.

(13)【答案】 2.

【解】 $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 的特征值为 $\lambda_1 = 3, \lambda_2 = \lambda_3 = 0$,

$\alpha\beta^T = \begin{pmatrix} 1 & 0 & k \\ 1 & 0 & k \\ 1 & 0 & k \end{pmatrix}$, 因为 $\alpha\beta^T$ 与 $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 相似, 所以 $\alpha\beta^T$ 的特征值也为 $\lambda_1 = 3, \lambda_2 = \lambda_3 = 0$,

再由 $\text{tr } \alpha\beta^T = 1 + k = \lambda_1 + \lambda_2 + \lambda_3 = 3$, 得 $k = 2$.

方法点评: 本题考查矩阵特征值性质与矩阵相似.

设 α, β 为 n 维列向量, 令 $A = \alpha\beta^T$, 则 $\text{tr } A = \alpha^T \beta = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.

(14)【答案】 np^2 .

【解】 因为总体 $X \sim B(n, p)$, 所以 $E(X) = np$, $D(X) = np(1-p)$,

又因为 $E(\bar{X}) = E(X) = np$, $E(S^2) = D(X) = np(1-p)$,

所以 $E(T) = E(\bar{X}) - E(S^2) = np - np(1-p) = np^2$.

三、解答题

(15)【解】 $f'_x(x, y) = 2x(2 + y^2)$, $f'_y(x, y) = 2x^2y + \ln y + 1$.

令 $\begin{cases} f'_x(x, y) = 2x(2 + y^2) = 0, \\ f'_y(x, y) = 2x^2y + \ln y + 1 = 0. \end{cases}$ 解得唯一驻点 $(0, \frac{1}{e})$.

因为 $A = f''_{xx}(0, \frac{1}{e}) = 2(2 + y^2) \Big|_{(0, \frac{1}{e})} = 2(2 + \frac{1}{e^2})$,

$B = f''_{xy}(0, \frac{1}{e}) = 4xy \Big|_{(0, \frac{1}{e})} = 0$,

$C = f''_{yy}(0, \frac{1}{e}) = (2x^2 + \frac{1}{y}) \Big|_{(0, \frac{1}{e})} = e$,

所以 $B^2 - AC = -2e(2 + \frac{1}{e^2}) < 0$, 又 $A > 0$, 于是 $(0, \frac{1}{e})$ 是 $f(x, y)$ 的极小值点, 极小

值为 $f(0, \frac{1}{e}) = -\frac{1}{e}$.

(16)【解】 令 $\sqrt{\frac{1+x}{x}} = t$, 则 $x = \frac{1}{t^2 - 1}$,

于是 $\int \ln\left(1 + \sqrt{\frac{1+x}{x}}\right) dx = \int \ln(1+t) d\left(\frac{1}{t^2-1}\right) = \frac{\ln(1+t)}{t^2-1} - \int \frac{1}{(t^2-1)(t+1)} dt$,

令 $\frac{1}{(t^2-1)(t+1)} = \frac{1}{(t+1)^2(t-1)} = \frac{A}{t-1} + \frac{B}{t+1} + \frac{C}{(t+1)^2}$,

解得 $A = \frac{1}{4}$, $B = -\frac{1}{4}$, $C = -\frac{1}{2}$,

则 $\int \frac{1}{(t^2-1)(t+1)} dt = \frac{1}{4} \int \left[\frac{1}{t-1} - \frac{1}{t+1} - \frac{2}{(t+1)^2} \right] dt$
 $= \frac{1}{4} \ln(t-1) - \frac{1}{4} \ln(t+1) + \frac{1}{2(t+1)} + C$,

故 $\int \ln\left(1 + \sqrt{\frac{1+x}{x}}\right) dx = \frac{\ln(1+t)}{t^2-1} + \frac{1}{4} \ln \frac{t+1}{t-1} - \frac{1}{2(t+1)} + C$
 $= x \ln\left(1 + \sqrt{\frac{1+x}{x}}\right) + \frac{1}{2} \ln(\sqrt{1+x} + \sqrt{x}) - \frac{1}{2} \frac{\sqrt{x}}{\sqrt{1+x} + \sqrt{x}} + C$
 $= x \ln\left(1 + \sqrt{\frac{1+x}{x}}\right) + \frac{1}{2} \ln(\sqrt{1+x} + \sqrt{x}) + \frac{1}{2} x - \frac{1}{2} \sqrt{x+x^2} + C$.

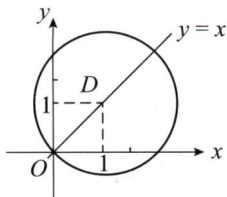
(17)【解】 方法一 如图所示(见下页), 令 $\begin{cases} x-1 = r \cos \theta, \\ y-1 = r \sin \theta, \end{cases}$

则 $D = \{(r, \theta) \mid \frac{\pi}{4} \leq \theta \leq \frac{5\pi}{4}, 0 \leq r \leq \sqrt{2}\}$,

$$\begin{aligned}\text{于是} \iint_D (x-y) dx dy &= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} d\theta \int_0^{\sqrt{2}} r^2 (\cos \theta - \sin \theta) dr = \frac{2\sqrt{2}}{3} \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\cos \theta - \sin \theta) d\theta \\ &= \frac{4}{3} \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \cos\left(\theta + \frac{\pi}{4}\right) d\left(\theta + \frac{\pi}{4}\right) = -\frac{8}{3}.\end{aligned}$$

方法二 令 $\begin{cases} x = r \cos \theta, \\ y = r \sin \theta \end{cases} \left(\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}, 0 \leq r \leq 2(\sin \theta + \cos \theta) \right),$

$$\begin{aligned}\text{则} \iint_D (x-y) dx dy &= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} d\theta \int_0^{2(\sin \theta + \cos \theta)} r^2 (\cos \theta - \sin \theta) dr \\ &= \frac{8}{3} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos \theta - \sin \theta) (\sin \theta + \cos \theta)^3 d\theta \\ &= \frac{8}{3} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\sin \theta + \cos \theta)^3 d(\sin \theta + \cos \theta) \\ &= \frac{2}{3} (\sin \theta + \cos \theta)^4 \Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}} = -\frac{8}{3}.\end{aligned}$$



三(17) 题图

(18)【证明】 (I) 取 $F(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$, 由题意知 $F(x)$ 在 $[a, b]$ 上连续,

在 (a, b) 内可导, 且 $F(a) = f(a) - \frac{f(b)-f(a)}{b-a}(a-a) = f(a)$,

$$F(b) = f(b) - \frac{f(b)-f(a)}{b-a}(b-a) = f(a).$$

根据罗尔定理, 存在 $\xi \in (a, b)$, 使得 $F'(\xi) = f'(\xi) - \frac{f(b)-f(a)}{b-a} = 0$, 即

$$f(b) - f(a) = f'(\xi)(b-a).$$

(II) 对于任意的 $t \in (0, \delta)$, 函数 $f(x)$ 在 $[0, t]$ 上连续, 在 $(0, t)$ 内可导, 由右导数定义及拉格朗日中值定理

$$f'_+(0) = \lim_{t \rightarrow 0^+} \frac{f(t) - f(0)}{t - 0} = \lim_{t \rightarrow 0^+} \frac{f'(\xi)t}{t} = \lim_{t \rightarrow 0^+} f'(\xi), \text{ 其中 } \xi \in (0, t)$$

由于 $\lim_{t \rightarrow 0^+} f'(t) = A$, 且当 $t \rightarrow 0^+$ 时, $\lim_{t \rightarrow 0^+} f'(\xi) = A$, 故 $f'_+(0)$ 存在, 且 $f'_+(0) = A$.

(19)【解】 曲线 $y = f(x)$ 与直线 $y=0, x=1$ 及 $x=t (t > 1)$ 所围成的曲边梯形绕 x 轴旋转

一周所得的立体的体积为 $V(t) = \pi \int_1^t f^2(x) dx$,

曲边梯形的面积为 $S(t) = \int_1^t f(x) dx$,

由题意得 $\pi \int_1^t f^2(x) dx = \pi t \int_1^t f(x) dx$, 即 $\int_1^t f^2(x) dx = t \int_1^t f(x) dx$, 两边求导数得

$$f^2(t) = \int_1^t f(x) dx + t f(t),$$

两边再求导数得

$$2f(t)f'(t) = 2f(t) + t f'(t),$$

则 $y = f(t)$ 满足的微分方程为 $(2y - t)y' = 2y$, 整理得 $\frac{dy}{y} + \frac{1}{2y}t = 1$,

$$\text{解得 } t = \left(\int e^{\frac{1}{2y} dy} dy + C \right) e^{-\int \frac{1}{2y} dy} = Cy^{-\frac{1}{2}} + \frac{2}{3}y,$$

$$\text{在 } f^2(t) = \int_1^t f(x) dx + tf(t) \text{ 中令 } t=1, \text{ 得 } f(1)=1, \text{ 代入 } t=Cy^{-\frac{1}{2}} + \frac{2}{3}y, \text{ 得 } C=\frac{1}{3},$$

$$\text{所求的曲线为 } x = \frac{1}{3} \left(\frac{1}{\sqrt{y}} + 2y \right).$$

(20)【解】(I)

$$\text{由 } (A \vdots \xi_1) = \left(\begin{array}{ccc|c} 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 0 & -4 & -2 & -2 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & -1 & -1 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$\text{得 } \xi_2 = k_1 \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix} + \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}k_1 - \frac{1}{2} \\ -\frac{1}{2}k_1 + \frac{1}{2} \\ k_1 \end{pmatrix} \quad (k_1 \text{ 为任意常数}).$$

$$A^2 = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ 0 & -4 & -2 \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ 0 & -4 & -2 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 0 \\ -2 & -2 & 0 \\ 4 & 4 & 0 \end{pmatrix},$$

$$\text{由 } (A^2 \vdots \xi_1) = \left(\begin{array}{ccc|c} 2 & 2 & 0 & -1 \\ -2 & -2 & 0 & 1 \\ 4 & 4 & 0 & -2 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$\text{得 } \xi_3 = k_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_2 - \frac{1}{2} \\ k_2 \\ k_3 \end{pmatrix} \quad (k_2, k_3 \text{ 为任意常数}).$$

(II) 方法一 由

$$|\xi_1, \xi_2, \xi_3| = \begin{vmatrix} -1 & \frac{1}{2}k_1 - \frac{1}{2} & -k_2 - \frac{1}{2} \\ 1 & -\frac{1}{2}k_1 + \frac{1}{2} & k_2 \\ -2 & k_1 & k_3 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -\frac{1}{2} \\ 1 & -\frac{1}{2}k_1 + \frac{1}{2} & k_2 \\ -2 & k_1 & k_3 \end{vmatrix} = -\frac{1}{2} \neq 0 \text{ 得}$$

ξ_1, ξ_2, ξ_3 线性无关.

方法二 设

$$k_1 \xi_1 + k_2 \xi_2 + k_3 \xi_3 = \mathbf{0}, \quad (1)$$

① 两边左乘 A 得 $k_1 A \xi_1 + k_2 A \xi_2 + k_3 A \xi_3 = \mathbf{0}$, 由 $A \xi_1 = \mathbf{0}$ 得

$$k_2 \xi_1 + k_3 A \xi_3 = \mathbf{0}, \quad (2)$$

② 两边左乘 A 得 $k_3 A^2 \xi_3 = \mathbf{0}$, 即 $k_3 \xi_1 = \mathbf{0}$.

由 $\xi_1 \neq \mathbf{0}$ 得 $k_3 = 0$, 代入 (2) 得 $k_2 = 0$, 再代入 (1) 得 $k_1 = 0$,

故 ξ_1, ξ_2, ξ_3 线性无关.

(21)【解】 (I) 二次型的矩阵 $A = \begin{pmatrix} a & 0 & 1 \\ 0 & a & -1 \\ 1 & -1 & a-1 \end{pmatrix}$,

由于 $|\lambda E - A| = \begin{vmatrix} \lambda - a & 0 & -1 \\ 0 & \lambda - a & 1 \\ -1 & 1 & \lambda - a + 1 \end{vmatrix} = (\lambda - a)[\lambda - (a + 1)][\lambda - (a - 2)]$,

所以 A 的特征值为 $\lambda_1 = a$, $\lambda_2 = a + 1$, $\lambda_3 = a - 2$.

(II) 方法一 由于 f 的规范形为 $y_1^2 + y_2^2$, 所以 A 合同于 $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 其秩为 2.

故 $|A| = \lambda_1 \lambda_2 \lambda_3$, 于是 $a = 0$ 或 $a = -1$ 或 $a = 2$.

当 $a = 0$ 时, $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = -2$, 此时 f 的规范形为 $y_1^2 - y_2^2$, 不合题意;

当 $a = -1$ 时, $\lambda_1 = -1$, $\lambda_2 = 0$, $\lambda_3 = -3$, 此时 f 的规范形为 $-y_1^2 - y_2^2$, 不合题意;

当 $a = 2$ 时, $\lambda_1 = 2$, $\lambda_2 = 3$, $\lambda_3 = 0$, 此时 f 的规范形为 $y_1^2 + y_2^2$.

综上所述, $a = 2$.

方法二 由于 f 的规范形为 $y_1^2 + y_2^2$, 所以 A 的特征值有 2 个为正数, 1 个为零.

又 $a - 2 < a < a + 1$, 所以 $a = 2$.

(22)【解】 (I) 随机变量 X 的边缘密度为 $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$,

当 $x \leq 0$ 时, $f_X(x) = 0$;

当 $x > 0$ 时, $f_X(x) = \int_0^x e^{-x} dy = x e^{-x}$,

则 $f_X(x) = \begin{cases} x e^{-x}, & x > 0, \\ 0, & x \leq 0, \end{cases}$

于是 $f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{1}{x}, & x > 0, 0 < y < x, \\ 0, & \text{其他.} \end{cases}$

(II) Y 的边缘密度为 $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$,

当 $y \leq 0$ 时, $f_Y(y) = 0$;

当 $y > 0$ 时, $f_Y(y) = \int_y^{+\infty} e^{-x} dx = e^{-y}$,

于是 $P\{X \leq 1 | Y \leq 1\} = \frac{P\{X \leq 1, Y \leq 1\}}{P\{Y \leq 1\}} = \frac{\iint_{x \leq 1, y \leq 1} f(x, y) dx dy}{\int_0^1 f_Y(y) dy}$,

而 $\iint_{x \leq 1, y \leq 1} f(x, y) dx dy = \int_0^1 dx \int_0^x e^{-x} dy = \int_0^1 x e^{-x} dx = 1 - \frac{2}{e}$,

$\int_0^1 f_Y(y) dy = \int_0^1 e^{-y} dy = 1 - \frac{1}{e}$,

故 $P\{X \leq 1 | Y \leq 1\} = \frac{e - 2}{e - 1}$.

$$(23) \text{【解】} \quad (\text{I}) \quad P\{X=1 \mid Z=0\} = \frac{P\{X=1, Z=0\}}{P\{Z=0\}} = \frac{C_2^1 \times C_2^1}{\frac{6 \times 6}{\left(\frac{1}{2}\right)^2}} = \frac{4}{9}.$$

(II) 由题意知 (X, Y) 的可能取值为 $(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (2, 0), (2, 1), (2, 2)$.

$$P\{X=0, Y=0\} = \frac{3^2}{6^2} = \frac{1}{4}, \quad P\{X=0, Y=1\} = \frac{C_2^1 \times C_2^1 \times C_3^1}{6^2} = \frac{1}{3},$$

$$P\{X=0, Y=2\} = \frac{2^2}{6^2} = \frac{1}{9}, \quad P\{X=1, Y=0\} = \frac{C_2^1 \times C_3^1}{6^2} = \frac{1}{6},$$

$$P\{X=1, Y=1\} = \frac{C_2^1 \times C_2^1}{6^2} = \frac{1}{9}, \quad P\{X=1, Y=2\} = 0,$$

$$P\{X=2, Y=0\} = \frac{1 \times 1}{6^2} = \frac{1}{36}, \quad P\{X=2, Y=1\} = 0, \quad P\{X=2, Y=2\} = 0,$$

X	Y		
	0	1	2
0	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{9}$
1	$\frac{1}{6}$	$\frac{1}{9}$	0
2	$\frac{1}{36}$	0	0